

P.5 SCIENCE NOTES TERM 1- 2024

TOPIC ONE

KEEPING POULTRY AND BEES

POULTRY KEEPING

Qn: What is poultry?

Poultry are all kinds of domestic birds or fowls.

Qn: What is poultry keeping?

Poultry keeping is the rearing of domestic birds or fowls.

Qn: Explain the following terms as used in poultry.

(a) A chick

A chick is a young chicken.

(b) A hen

A hen is a female chicken.

(c) A cock

A cock is a male chicken.

(d) A pullet

A pullet is a young female chicken.

(e) A cockerel

A cockerel is a young male chicken.

(f) A capon

A capon is a castrated cock.

(e) Culling

Culling is the removal of sick and unproductive birds from the flock.

(h) Candling

Is the viewing of an egg in bright light to know if it is fertilized or not.

(i) Incubation

Incubation is the providing of necessary conditions to a fertilized egg to

enable it hatch .

(j) Incubator

Is a machine that provides necessary conditions to eggs until they hatch .

(k) Brooding

This is the providing of special care to chicks below 8 weeks.

(l) A brooder

Is a special structure where chicks below 8 weeks are kept.

(m) A broody hen

Is a hen incubating its eggs.

(n) Preening

Is the act of birds cleaning and arranging their feathers.

(o) Moulting

Is the shedding of old feathers from the body of a bird.

(p) Debeaking

Is the shortening of the upper beak of a bird.

Qn: Give any two reasons why people keep poultry.

To get eggs

To get meat

To sell and get money

To get manure

Qn: Identify any two products got from poultry.

Eggs

Meat

Feathers

Manure

Qn: Mention any three types of poultry / Examples of poultry.

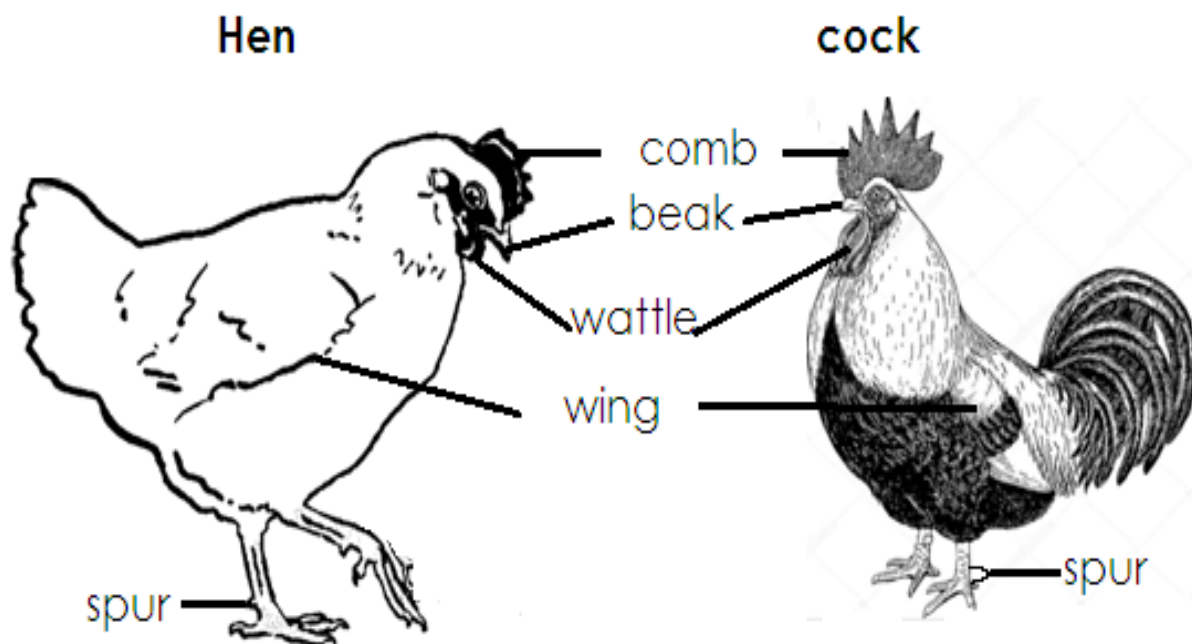
Chicken

Guinea fowl

Geese

pigeon

features of domestic fowls. External



Functions of some parts

(a) Beak

For protection

For pecking food

For building nests

(b) Comb and wattle

To regulate bird's body temperature .

To attract mates.

(c) Spur

For protection

(d) Feathers

To keep the body of a bird warm.

For flying

For identification

For incubating eggs

For brooding chicks

They give shape to the bird.

(e) Toes and claws

For scratching the ground when looking for food.

For protection.

Qn: Give any two structural differences between a hen and a cock.

A hen has a small comb while a cock has a big comb.

A hen has a small wattle while a cock has a big wattle.

A hen has dull short neck feathers while a cock has bright and long neck and tail feathers.

A hen has small earlobes while a cock has big earlobes.

A hen has a short spur while a cock has a long spur.

FEATHERS

Qn: Mention any four uses of feathers to a bird.

Types of feathers.

(i) Quill or flight feathers.

- (ii) Body or covert feathers
- (iii) Down feathers
- (iv) Filoplume feathers/ hair feathers.

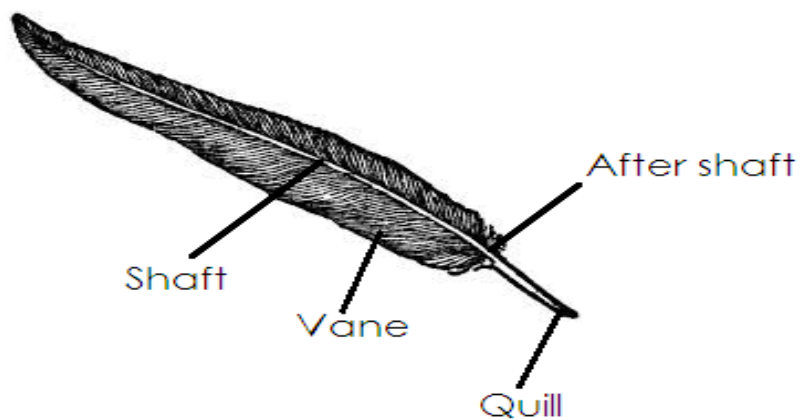
QUILL FEATHERS

Qn: Mention two body parts of a bird where quill feathers are found.

Wings

Tail

Illustration showing the quill feather.

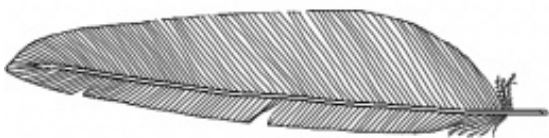


BODY / COVERT FEATHERS

They keep the bird warm.

They give the bird

Illustration showing the body feather.



DOWN FEATHERS

These are feathers that the chick is hatched with.

They keep the chick warm.



FILOPLUME FEATHERS

They are the smallest feathers found nearest to the skin of a bird.

They are seen after removing the body and down feathers.

Illustration showing a filoplume feather



BREEDS OF CHICKEN

Qn: What is a breed of chicken?

A breed of chicken is a class of chicken having specific characteristics.

Types of breeds of chicken

Local breeds

Exotic breeds

Cross breeds/Hybrids

Local breeds of chicken

These are chicken that exist in a country without being brought from outside countries.

Qn: State any three characteristics of local breeds of chicken.

(i) They are more resistant to diseases.

(ii) They are more resistant to harsh weather changes.

(iii) They grow slowly

(iv) They lay fewer eggs.

Exotic breeds of chicken

These are chicken that have been imported into E. African from outside countries.

Qn: Write down any three characteristics of exotic breeds of chicken.

(i) They are less resistant to diseases.

(ii) They are less resistant to harsh weather changes

(iii) They grow quickly

(iv) They have the same colour.

(v) They have the small size.

(vi) They provide high quantity of products.

Qn: Give three examples of exotic breeds of chicken.

White leghorn

Brown egger

Rhode Island red

Minorca

Light Sussex

Qn: Identify three ways of improving on the local breeds of chicken.

(i) By cross breeding

(ii) By carrying out selective breeding

iii) By proper feeding

Qn: State any two importance of crossbreeding.

It improves on the quality of birds.

It improves on the productivity of birds.

It encourages fast growth in birds.

TYPES OF CHICKEN

Qn: What is a type of chicken?

A type of chicken is a class of chicken kept for a specific purpose.

These are three types chicken namely:-

- (i) Layers or light breeds
- (ii) Broilers or table birds
- (iii) Dual purpose chicken.

LAYERS

These are chicken that are mainly kept for egg production.

Qn: Write down any two examples of layers or light birds.

- (i) White leghorn
- (ii) Ancona
- (iii) Minorca
- (iv) Brown eggers
- (v) Sykes

BROILERS

These are chicken that are mainly kept from meat production.

Examples of broilers birds.

- (i) Black australorp

- (ii) Plymouth rock
- (iii) Newhampshire
- (iv) Rhode Island Red
- (v) Light Sussex

Dual purpose Birds

These are chicken kept for both meat and egg production.

Examples of dual purpose chicken

Ross birds

Shavers

SYSTEM OF KEEPING POULTRY

There are four systems of keeping poultry namely.

- (a) Free range system
- (b) Deep litter system
- (c) Battery / cage system
- (d) Fold / pen system

1. FREE RANGE SYSTEM

This is a system where birds are left to move about and look for their food.

Advantages of free range system

- (i) It is cheap in terms of feeding.
- (ii) Birds get enough physical exercise
- (iii) Birds eat food of their choice
- (iv) Birds feed on a balanced diet.
- (v) The farmer gets time to do other activities.
- (vi) Poultry vices are reduced.

Disadvantages of free-range system

- (i) Birds can easily get lost.
- (ii) Birds can easily get stolen
- (iii) Birds can easily be eaten by wild animals.
- (iv) Birds can destroy people's property.
- (v) It is not easy to collect eggs.
- (vi) It is not easy to control diseases.

2. DEEP LITTER SYSTEM

This is a system where birds are kept in a house all the time.

Advantages of deep litter system

- (i) Birds cannot easily get lost.
- (ii) Birds cannot easily be stolen.
- (iii) Birds cannot easily be eaten by wild animals.
- (iv) Birds are protected from bad weather changes.
- (v) It is easy to identify a sick bird.

Disadvantages of deep litter system

- (i) Birds do not get enough physical exercise.
- (ii) It is expensive
- (iii) Birds do not eat food of their choice.
- (iv) It is not easily to control poultry vices.

Qn: What is litter?

Litter is material put on the floor of a poultry house.

Qn: State any two importance of litter in a poultry house.

- (i) Litter provides warmth to poultry.
- (ii) Litter prevents eggs from breaking.

(iii) Litter absorbs moisture from poultry droppings.

Qn: Why should litter be changed once it is old?

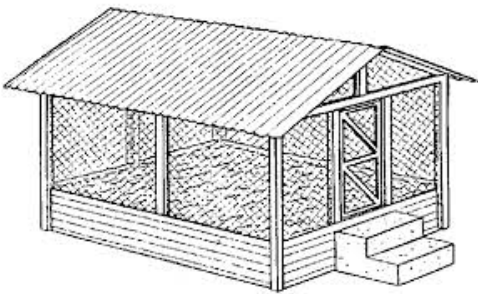
To prevent outbreak of diseases.

To control parasites.

Qn. How does litter prevent eggs from breaking?

Litter provide a soft landing of eggs.

Diagram showing deep litter system.



Qn: State the importance of the following in a poultry house.

(a) Perches

They control boredom in birds.

They act as resting places for birds.

They enable poultry to carry out physical exercises.

(b) A feeding trough

It is where food for poultry is put.

(c) A conical drinking waterer

It is where drinking water for poultry is put.

3. Battery / cage system.

This is a system where birds are kept in separate cages.

Qn: Write down at least four advantages of battery/ cage system.

It is easy to collect eggs.

Birds cannot easily be stolen.

Birds cannot easily get lost.

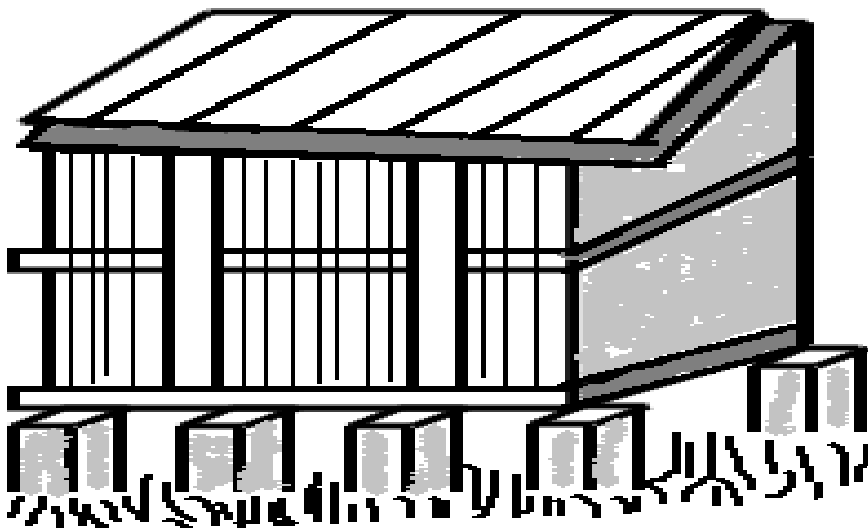
It is easy to control poultry vices.

It is easy to identify sick birds.

Disadvantages of battery / cage system

- (i) Birds do not get enough exercises.
- (ii) Birds do not eat food of their choice.
- (iii) It is expensive.

Diagram showing battery / cage system



4. Fold or pen system

This is a system where birds are kept in small houses called folds or pens.

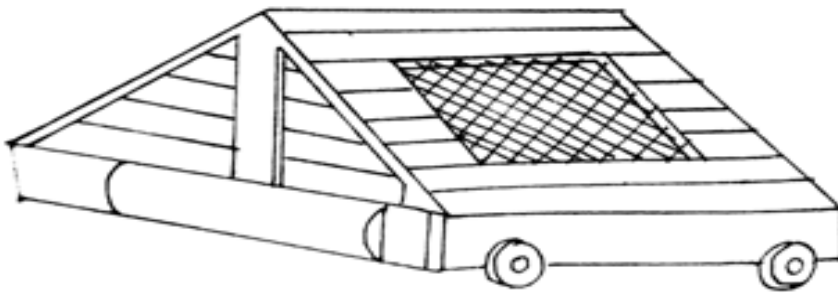
Advantages of fold / pen system.

- (i) Birds are protected from wild animals.
- (ii) Birds cannot easily get lost.
- (iii) Poultry feeds are not wasted.
- (iv) It is cheap compared to battery / cage system.
- (v) The farmer gets time to do other activities.

Disadvantages of fold / pen system.

- (i) Few birds are kept.
- (ii) Birds do not get enough exercise.
- (iii) Birds do not eat food of their choice.
- (v) It is tiresome to move the pens.

A diagram of a fold / pen system



FEEDING CHICKEN

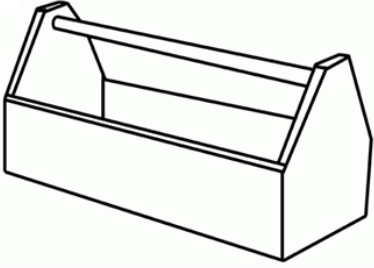
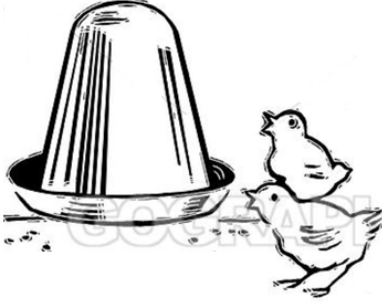
Qn: Give two importance of feeding chicken.

- (i) To enable them grow.
- (ii) To enable birds lay many eggs.

Qn: Mention the type of feeds given to the following groups of birds.

- | | | |
|----------------------------------|---|---------------|
| (a) Chicks | - | Chick mash |
| (b) Layers | - | Layers mash |
| (c) Broilers | - | Broilers mash |
| (d) Layers between 4 – 16 weeks | - | Growers mash |
| (e) Broilers between 4 – 8 weeks | - | Growers mash |

Qn: Name the items drawn below.

	
A feeding trough	Conical drinking waterer

Qn: State the importance of a swivel bar on a feeding trough.

A swivel bar rolls to prevent birds from stepping in feeds.

Give the function of a water trough in a poultry house.

It is where birds drink water from.

Qn: Mention the minerals salts that make the bird's egg shells and bone strong.

Calcium

Phosphorous

Qn: State the importance of calcium in poultry feeds.

Calcium enable birds lay hard shelled eggs.

Qn: Why do poultry farmers hang greens in a poultry house?

To enable birds to carry out physical exercises.

Qn: State the importance of greens to poultry.

Greens are sources of vitamins and minerals salts.

Practices done in poultry management.

(i) Regular cleaning of the house.

(ii) Culling

(iii) Debeaking

- (iv) Vaccination
- (v) Regular feeding
- (vi) Egg collection

POULTRY PARASITES

Qn: What are parasites?

Parasites are living organisms that depend on other organisms for food.

Types of parasites.

- (i) Ecto parasites/external parasites
- (ii) Endo parasites/internal parasites

Ecto parasites

These are parasites that live outside on the body of a bird.

- (i) Lice
- (ii) mites
- (iii) Fleas

(b) Endo parasites

These are parasites that live inside the body of a bird.

Examples of endo parasites

- (i) Tape worms
- (ii) Hook worms

Qn: State any three ways of controlling parasites in poultry.

- (i) Regular deworming
- (ii) Dust the body of a bird using chemicals.
- (iii) Regular cleaning of the poultry house.

INCUBATION

Qn: What is incubation?

This is the providing of necessary conditions to a fertilized egg to enable it hatch into a young one of a bird.

Good conditions for eggs to hatch.

- (i) Presence of optimum temperature.
- (iii) Presence of oxygen

Qn: Write down any three factors that can make a fertilized egg fail to hatch.

- (i) When the egg is too dirty.
- (ii) When the egg has a soft shell.
- (iii) When the egg has two yolks.
- (iv) When the egg does not have air space.
- (vi) Over heating during incubation

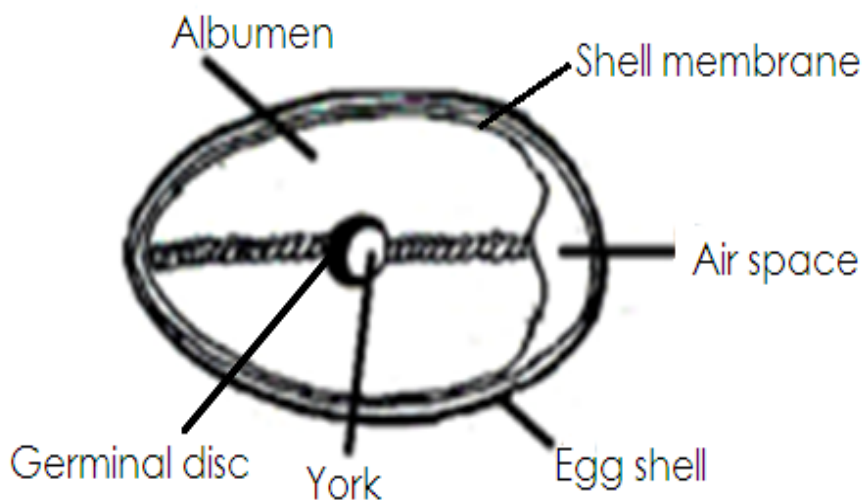
Qn: What is incubation period?

Is the time taken for an egg to hatch into a a young one of a bird.

Qn: State the incubation period of each of the following birds.

- (a) Hen - 21 days
- (b) Pigeon - 14 days
- (c) Geese - 30 days
- (d) Turkey - 28 days
- (e) Duck - 30 – 31 days

Structure showing parts of an egg.



Function of each part.

(a) Egg shell.

Protects the inside parts of the egg.

(b) Air space

Provides oxygen to the embryo.

It takes out carbon dioxide which is given off by the embryo.

(c) Chalaza/twisted chalaza.

It holds the Yolk and embryo in position.

It carries food and water to the embryo.

It carries oxygen from the air space to the embryo.

(d) Albumen / egg white

It is a source of proteins to the embryo.

(e) Egg yolk

It provides food for the embryo.

(f) Germinal disc

Develops into a young bird.

Qn: Why is the egg shell porous?

To allow gaseous exchange to take place easily.

TYPES OF INCUBATION

(i) Natural incubation

(ii) Artificial incubation

Natural incubation

Is the type of incubation where a hen provides necessary conditions to eggs in order to hatch into chicks.

Qn. State the advantages of natural incubation

It is cheap.

It does not require skilled knowledge.

Eggs are turned by the broody hen.

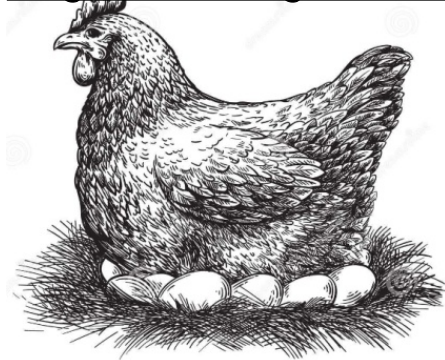
Disadvantages of natural incubation

Few eggs are hatched.

Eggs may not hatch in case the mother hen dies.

The hen can be easily attacked by parasites.

Diagram showing natural incubation



Artificial incubation

Is the type of incubation where an incubator provides necessary conditions to eggs until they hatch into chicks.

Advantages of artificial incubation

Many eggs are hatched at once.

An incubator can be used at any time.

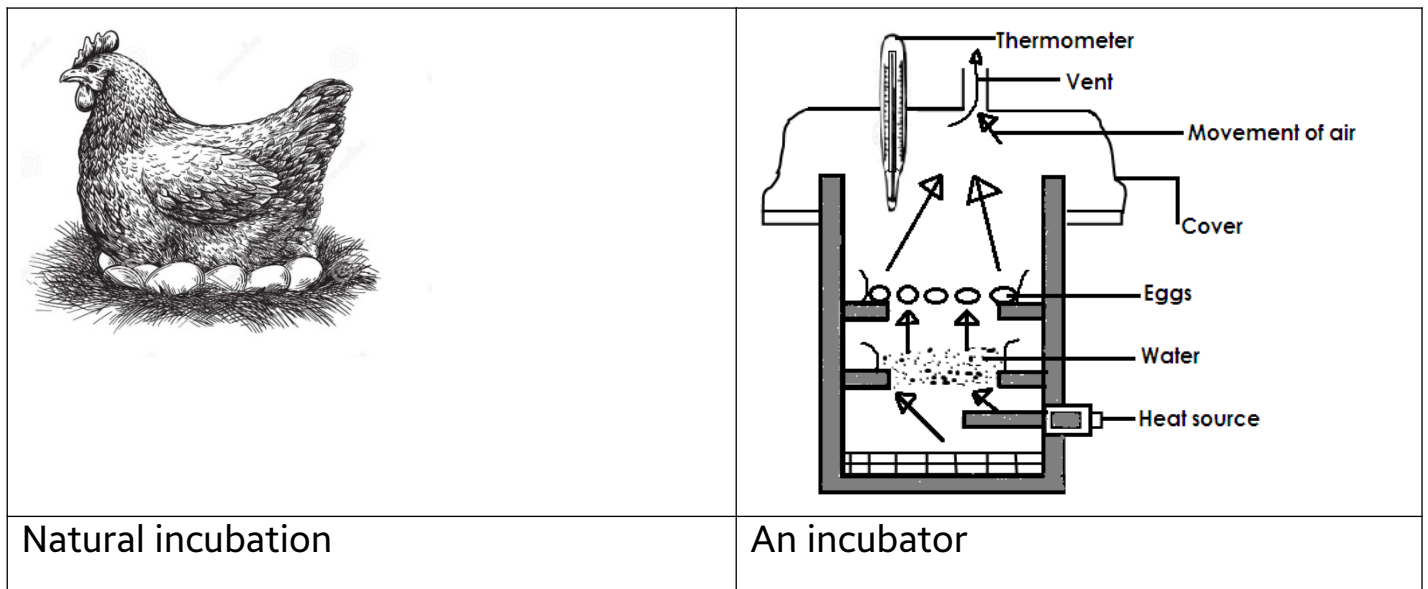
Disadvantages of artificial incubation

It is expensive.

It requires skilled labour.

It needs much care and attention.

Diagram showing an incubator



BROODING

Brooding is the special care given to chicks below 8 weeks.

TYPES OF BROODING

Natural brooding

Artificial brooding

NATURAL BROODING

Qn: What is natural brooding?

Is the type of brooding where the mother hen provides care to its chicks.

Advantages of natural brooding

(i) It is cheap

(ii) It does not need a lot of care and attention.

(iii) A farmer gets time to do other activities.

Disadvantages of natural brooding

(i) Chicks may be eaten by wild animals.

(ii) Chicks can easily be stolen.

(iii) Chicks can easily get lost.

(iv) Chicks can be easily affected by harsh weather like heavy rainfall.

Qn: Mention any two things a broody hen provides to her chicks.

Warmth

Food

Security

ARTIFICIAL BROODING

QnWhat is artificial brooding?

Is the type of brooding where chicks are kept in a special structure called a brooder.

Advantages of artificial brooding

Many chicks are kept in one place.

Chicks cannot easily be eaten by wild animals.

Chicks cannot easily be stolen.

Chicks cannot easily get lost.

Chicks are protected from bad weather.

It is easy to feed chicks in one place

Disadvantages of artificial brooding

It is expensive

Chicks can die in case heat is not enough.

It needs a lot of care and attention

Toe pecking among chicks is difficult to control.

Qn: Define a brooder.

A brooder is a special structure where chicks are reared.

Qn: State any two things provided to chicks in a brooder.

Warmth

Food

Water

Security

Types of brooders

(a) Charcoal brooders

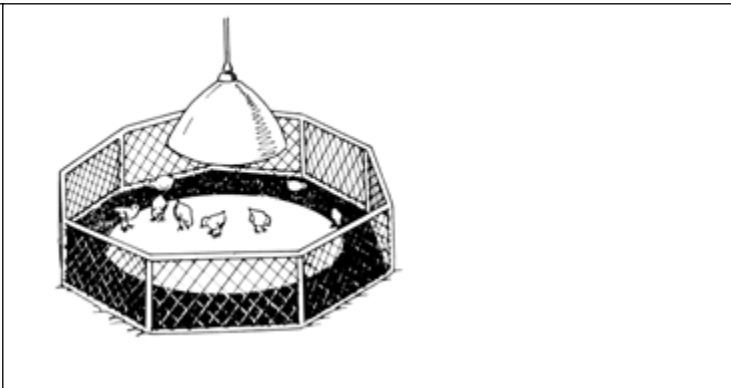
(b) Infrared lamp brooders

(c) Kerosene brooders

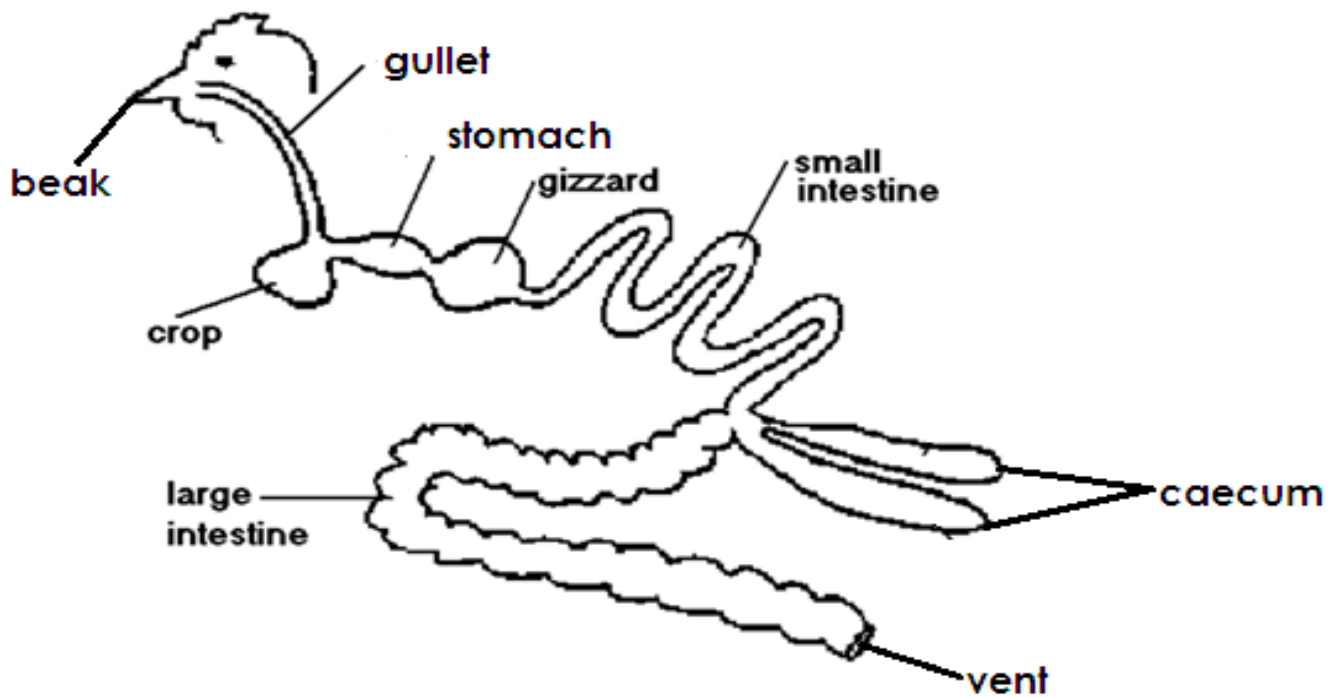
Qn: State any two sources of heat in a brooder.

Charcoal stove

Kerosene brooders.

	
Natural brooding	Artificial brooding.

The digestive system of a domestic bird.



Uses of each part.

(a) The beak

For pecking food.

(b) Gullet

It is a passage of food from mouth to the crop.

(c) Crop

It softens and moistens food.

It stores food for some time.

(d) Stomach

It is where food is mixed with digestive enzymes.

(e) Gizzard

It contains grit which crush food into small particles.

(f) Small intestines

It is where absorption of digested food takes place.

(g) Large intestines

It is where absorption of water takes place.

(h) Caeca

It stores waste materials before they are passed out.

(g) Large intestine

It is where absorption of water takes place.

(i) Vent

It is a passage of waste materials.

Qn: How is the nostril useful to a bird?

It is used for smelling.

Qn: State what happens to the food while in the crop of a bird.

It is softened.

It is moistened

Qn: Give the function of grits found in the gizzard of a bird.

For crushing food.

POULTRY DISEASES

Qn: Give any two causes of diseases in poultry.

Poor feeding

Poor hygiene in a poultry house.

Overcrowding of birds

Poor housing

Feeding birds with contaminated food and water.

Examples of poultry diseases and their causes.

Disease	Cause
Coccidiosis	Protozoa
Black head	Protozoa
Fowl typhoid	bacteria
Pneumonia	bacteria / virus
Fowl pox	Virus
New castle disease	Virus
Gumboro	Virus
Avian leucosis	virus

Poultry diseases, their prevention and treatment

Disease	Signs and symptoms	Prevention and treatment
Pneumonia	Coughing Mucus comes out of the nose Dullness Difficulty breathing	Keep the house clean. Make the house well ventilated. Cull the sick birds
Coccidiosis	Dropping wings Rough feathers Yellowish diarrhoea Droppings with blood Dropping wings.	Keep the house clean. Cull the sick birds
Fowl typhoid	Yellowish – green diarrhea	Burn or bury dead birds.

	Rough feathers Sleepy eyes Dullness	Cull the sick birds Vaccinate the birds.
New castle disease	Mucus from mouth and nose. Coughing Lameness Sudden death of birds in large numbers.	Keep the house clean. Vaccinate Kill and bury infected birds.
Fowl pox	Discharge of liquids from eyes and nostrils. Tiny wounds on the comb, wattle, wings and mouth. Eyes get sleepy and stuck.	Regular vaccination. Keep the house clean. Cull all sick birds.
Gumboro disease	Dropping wings. Diarrhoea with blood stains. Rough feathers Birds peck their feet.	Cull the sick birds. Vaccinate Burn or bury dead birds.

Qn: Write down any three ways of preventing and controlling disease in poultry.

- (i) Keep the poultry house clean.
- (ii) Regular vaccination
- (iii) Put coccidiostats in poultry feeds.
- (iv) Culling all sick birds

(v) Burn or burry dead birds.

Qn: List down any four effects of diseases and parasites on a poultry farm.

(i) They lead to stunted growth.

(ii) They reduce production

(iii) They reduce profits

(iv) They reduce income.

POULTRY VICES

Qn: What are poultry vices?

(i) Poultry vices are bad habits in poultry.

Qn: Write down any two causes of poultry vices.

(i) Hunger

(ii) Poor feeding

(iii) Overcrowding birds

(iv) Boredom

(v) Too much light in a poultry house.

(vi) Lack of calcium in poultry feeds.

(vii) Failure to collect eggs in time.

Examples of poultry vices

Cannibalism

Egg eating

Feather pecking

CANNIBALISM

It is the act of a bird eating another bird.

Causes of cannibalism

Prolapse

Crowding in the poultry house.

Lack of proteins in poultry feeds.

Starving birds

Prevention of cannibalism

Hanging enough green vegetables in the poultry house.

Regular feeding of birds.

Providing enough space for birds.

Avoiding overstocking

EGG EATING

- It is the act of birds eating their own eggs.

Causes of egg eating.

Hunger

Boredom

Too much light in the nesting area.

Lack of enough calcium in their bodies.

Failure by the farmer to collect eggs in time.

Presence of broken eggs.

Control of egg eating

De-beaking the birds

Hanging green vegetables.

Nesting places should be kept dark areas.

Provide feeds which contain calcium.

Collect eggs in time .

Qn. How does debeaking help to control egg eating?

It makes the beak of a bird blunt.

Qn. Mention any two devices used for debeaking.

Hot knife

Sharp pair of scissors

FARM RECORDS

Qn: What are farm records?

Farm records are written information about various activities carried out on a farm.

Qn: What is record keeping?

This is the keeping of information about various activities carried out on a farm.

Types of farm records

- (i) Health records
- (ii) Flock records
- (iii) Feed records
- (iv) Production records
- (v) Sales and expenses records.

Importance of farm records.

- (i) Farm records enable the farmer to plan for the farm.
- (ii) Farm records enable the farmer to know the history of the farm.
- (iii) Farm records help a farmer to know his income and expenditure.
- (iv) Farm records help farmer to know if he is making profits or losses.

BEE KEEPING

Terms used

(a) Apiculture

Is the rearing of honey bees.

(b) An a piarist

Is a bee farmer

(c) An apiary

Is a place where we find many bee hives.

(d) A colony

Is a large group of bees in a hive.

(e) A swarm

A swarm is large group of bees moving in the same direction.

(f) Swarming

This is the movement of bees from one place to another.

(g) A nuptial / maiden / wedding flight

This is the flight during which a drone bee mates with a queen bee.

(h) Stocking the bee hive

Is the encouraging of bees to occupy an empty bee hive.

(i) Setting or setting a bee hive

Is the selecting of a suitable place in which to put a bee hive.

Social insects

These are insects which move, live and work together in large groups.

Qn: Write down any examples of social insects

Honey bees

Wasps

Termites

Red ants

White ants

Black ants

Qn: What are solitary insects?

These are insects that do not live, move and work together.

Qn: Identify any two examples of solitary insects.

Houseflies

Mosquitoes

Moths

Cockroaches

Tsetse flies

Bumble bees

Types of honey bees / castes of honey bees.

Queen bee

Drone bee

Worker bees

QUEEN BEE

It is the largest in a bee hive.

It has a longer abdomen

It is a female bee.

It lays eggs.

It has an ovipositor for laying eggs.

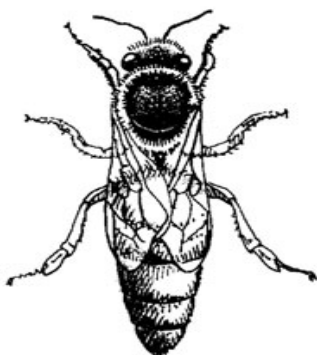
It feeds of a special food called royal jelly.

It develops from the fertilized eggs.

Qn: State the role of a queen bee in the hive.

To lay eggs

Structure of a queen bee.



DRONE BEES

They are the male bees in a hive.

Drones develop from unfertilized eggs.

Drones do not have a sting.

They make a buzzing sound while flying.

Qn: State the role of a drone bee

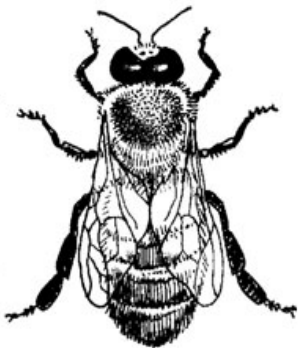
To mate with the queen bee.

Qn: Why does a drone bee die shortly after mating with the queen?

The sexual organ breaks off during mating hence damaging the abdomen.

The abdomen gets damaged after mating.

Structure of a drone bee.



WORKER BEES

They are female sterile bees in a hive.

They are smallest in the hive.

They have pollen basket on their hind legs

They develop from the fertilized eggs.

They do not have ovipositor.

They have a sting.

Duties of worker bee.

They build and repair the hive.

They collect pollen and propolis.

They protect the hive.

They feed the queen.

They feed the grub.

They defend the hive.

They produce wax.

They make honey.

Structure of a worker bee.



Qn: Why is a worker bee unable to lay eggs?

It does not have an ovipositor.

Qn: Why are worker bees called female sterile bees?

They do not lay eggs.

Qn. Why does a worker bee die after stinging a person?

The loss of sting results in damaging of the abdomen.

The abdomen gets damaged.

Qn. How do worker bees control temperature in a hive?

By rapid flapping of their wings.

Qn. Name the food given to the queen bee by the worker bee.

Royal jelly

Qn. Write down any two things collected by bees from the environment.

Nectar

Propolis

Pollen

Water

Qn. Why do worker bees collect water and nectar?

For making honey.

Qn. State the use of propolis to bees?

For repairing the hive.

For smoothing the inside of the bee hive.

Water proofs the hive.

Qn. Name the part of a worker bee that produces wax.

Wax glands

Importance of bees to people.

People get honey

People get bee wax from bees.

Bees pollinate flowers.

Importance of bees to plants.

Bees help in pollination of plants.

Products got from bees.

Honey

Bee wax

Qn. Give any two uses of honey.

Honey is used as medicine.

Honey is eaten as food.

Honey is used to sweeten bread.

Honey is used to sweeten tea.

Honey is a source of income when sold.

Qn. What food value is obtained from eating honey?

Carbohydrates.

Uses of bee wax

(i) Bee wax is used to make candle wax.

(ii) Bee wax is used to make shoe polish.

(iii) Bee wax is used to make cosmetics.

(iv) Bee wax is used to make crayons.

Products obtained from bee wax

Candle wax

Shoe polish

Floor polish

Cosmetics

LIFE CYCLE OF A HONEY BEE

Qn. How do bees reproduce?

By laying eggs.

Qn. Which type of life cycle do honey bees undergo?

Complete life cycle.

Qn. Name the stages of development a honey bee undergo.

Eggs

Larva

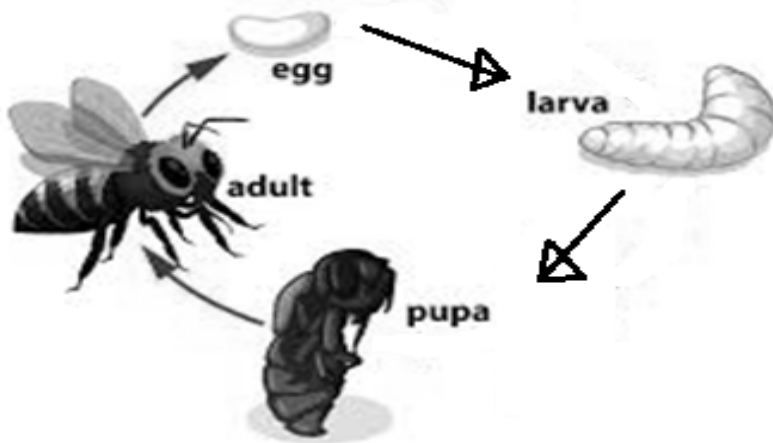
Pupa

Adult

Qn. What name is given to the larva stage of a honey bee?

Grub

Illustration showing the life cycle of a honey bee.



SWARMING

Swarming is the movement of bees from one place to another looking for a new hive.

Causes of swarming in bees.

- Direct sunlight to the hive.
- Direct smoke into the hive.
- Too much noise around the bee hive.
- Bad smell in the bee hive.
- When the queen bee hives.
- When another queen is hatched.
- Leaking bee hive.
- Shortage of flowers and water in a place.
- Overcrowding of bees in a hive.
- Attacks from their enemies.

Bad smell around the hive.

ENEMIES OF BEES

Honey badgers

Safari ants

Wax moths

Termites

Hive beetles

Ways of preventing and controlling bee enemies

- (i) Smearing oil on poles of bee hives.
- (ii) Spraying using insecticides.
- (iii) Spraying using insecticides.
- (iv) Hang bee hives between two poles.
- (v) Keep grass around bee hives short.

TYPES OF BEE HIVES

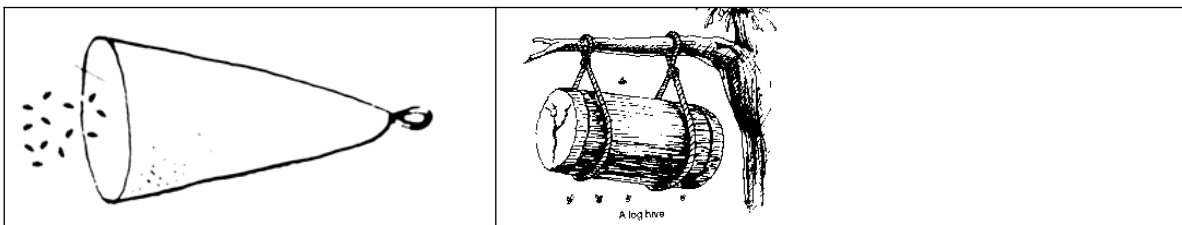
- (a) Traditional bee hives.
- (b) Modern bee hive.

Traditional bee hives

These are hives made from local materials.

Examples of traditional bee hives

- (i) Kigezi bee hive
- (ii) Dugout log hive.
- (iii) Tin hive



Kigezi hive	Dug out lo hive
-------------	-----------------

Advantages of traditional bee hives.

- (i) They are cheap and easy to make.
- (ii) The colony is not disturbed by the bee keeper.

Disadvantages of a traditional hive.

- (i) The hive is damaged in the process of harvesting honey.
- (ii) The honey harvested is not always clean.
- (iii) The hive does not last for long.
- (iv) It is not easy to inspect the colony.

MODERN BEE HIVES

Examples of modern bee hives

- (a) Box hive
- (b) Top bar hive

Parts of a modern bee hive

(a) Queen excluder

It prevents the queen from going to the honey chamber.

(b) The holes/ entrance.

it is a passage of bees into hive.

(c) Honey chamber

It is where honey is stored.

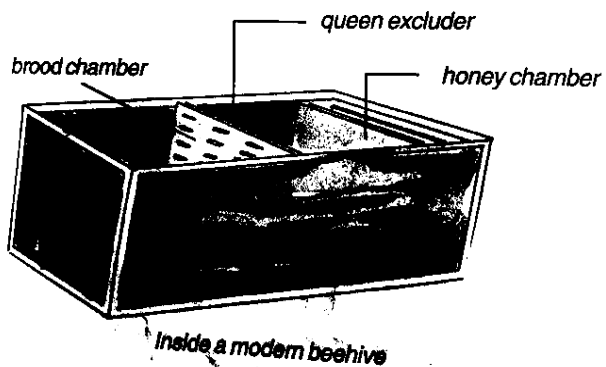
(d) Brood chamber

It is where the queen bee lays her eggs.

Qn. Why is the excluder made of tiny holes?

To allow only the worker bees which have to prepare honey to enter the honey chamber.

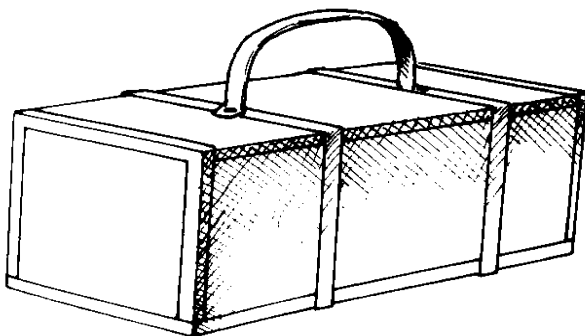
Inside a modern bee hive.



A diagram of a top bar hive



Box hive



Advantages of a modern hive

Clean honey is harvested.

It is easy to inspect the colony.

A modern bee hive lasts for a long time.

A hive is not destroyed during honey harvested.

The colony develops undisturbed.

Disadvantages of a modern hive.

It is expensive to make.

The colony is disturbed by the bee keeper.

Sitting the hive

This is the selection of a suitable place in which to put bee hives.

Factors to consider when selecting a suitable place in which to put bee hives.

- (i) The place should be free from noise.
- (ii) The place should be away from the main road.
- (iii) The place should be near flowering plants.
- (iv) The place should be near a water source.
- (v) The place should be far from people and animals.
- (vi) The place should be sheltered from direct sunlight.

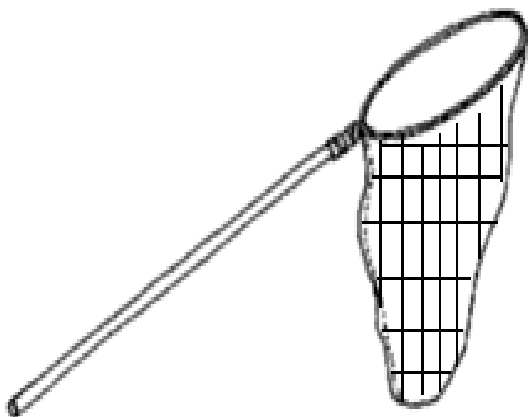
STOCKING THE HIVE

Is the act of encouraging bees to occupy an empty hive.

Qn. Give any two ways of stocking a bee hive.

- (i) By smearing bee wax in a hive.
- (ii) By using a swarm catcher or catcher box.
- (iv) By using a swarm catching net.

A swarm catching net.



HARVESTING HONEY

Qn. Equipments needed by a honey harvester.

Bucket - bee veil - Gumboot - gloves - smoker
knife

Harvesting honey is removing honey combs from the bee hive.

Methods of harvesting honey

(a) Traditional method.

(b) Modern method

Qn. State the function of each of the following equipment when harvesting honey.

(a) Bucket

It is where harvested honey is put.

(b) Knife

For cutting honey combs.

(c) Overall

It protects the body of the honey harvester from bee stings.

(d) Bee veil

Protects the head of the honey harvester from bee stings.

(e) Gloves

Protects the hands of the honey harvester from bee stings.

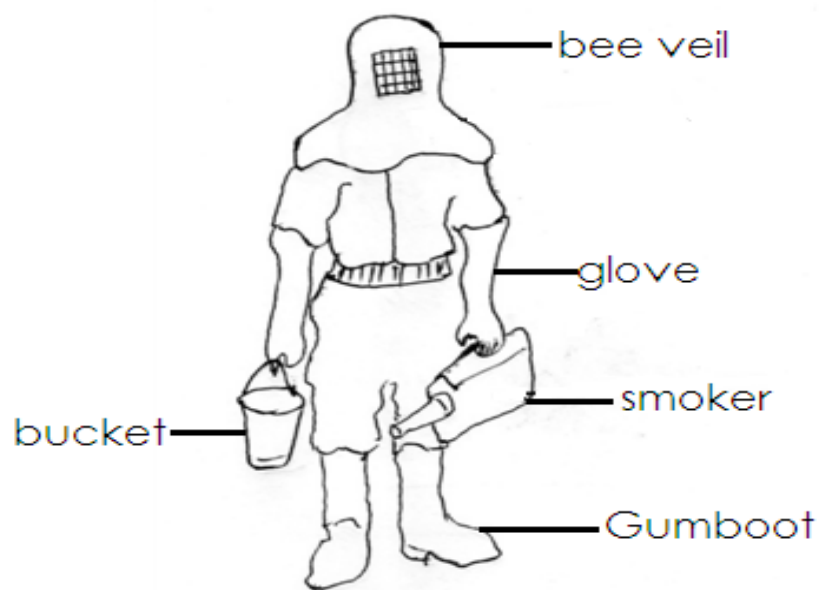
(f) Gum boots

Protect the feet of the honey harvester from bee stings.

(g) Smoker

Produces smoke that calms bees.

Diagram showing a honey harvester



Qn. Why is harvesting of honey best done in the evening?

All bees are in the hive and in active.

EXTRACTION OF HONEY

Extracting honey means removing honey from the honey combs.

Methods of extracting honey

- (a) Floating the wax method.
- (b) Pressing the honey method.
- (c) Centrifuging method.

Floating the wax method

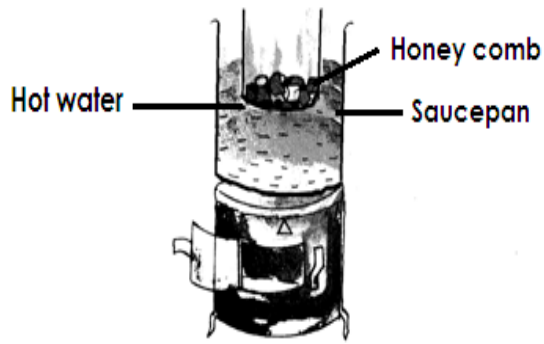
Break the honey combs and put them in a large container.

Put some water in a big sauce pan and place it on fire.

Place the container with honey combs on a big saucepan with boiling water.

The wax and honey will melt and wax will float on honey.

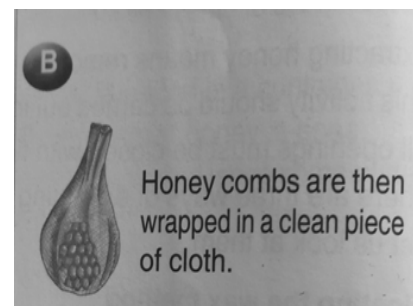
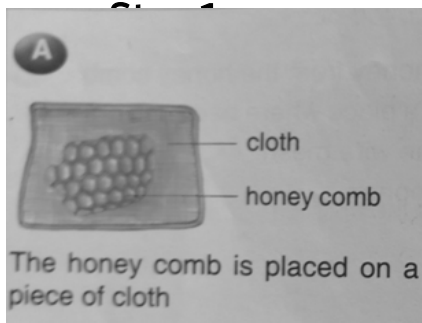
An illustration showing floating wax.



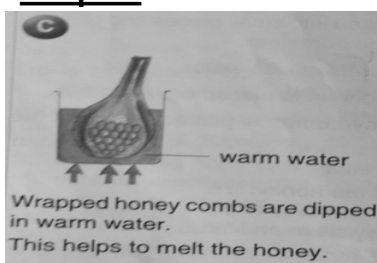
Pressing the honey method.

- (i) The honey comb is placed on a piece of cloth.
- (ii) Honey combs are then wrapped in a clean piece of cloth.
- (iii) Dip the wrapped honey combs in warm water.
- (iv) Squeeze the honey combs wrapped in a cloth to make honey come out.

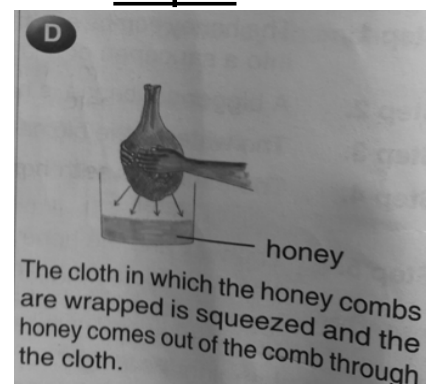
Illustration showing steps for pressing the honey method.



Step III



Step IV



Centrifuging method / spinning method

This is where a mechanic is used to extract honey from the honey combs.

Qn. What name is given to a machine used to extract honey from the honey combs?

Centrifugal honey extractor.

Steps:

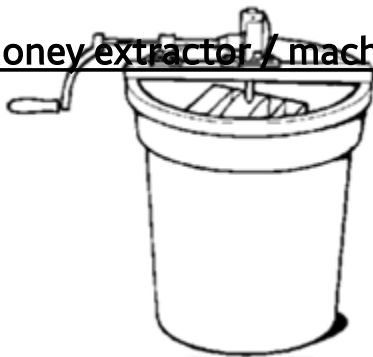
Remove the wax that seals the comb and put the honey comb in the machine.

The machine spins the honey combs round at a very high speed forcing honey to filter out.

The honey then settles at the bottom of the machine.

Boil the honey and store it a clean container.

A centrifugal honey extractor / machine.



Qn. State the importance of warm water during floating the honey method.

To make wax and honey melt.

Qn. Why can't honey go bad?

Honey has a lot of sugars and less water.

TOPIC TWO
MEASUREMENTS

Qn. What is a measurement?

Is the process of finding how long, short, small, big, heavy or light an object is.

Examples of measurements quantities.

Mass	- capacity	- Time	- volume	- Temperature
Density	- speed	- weight	- Distance	

Measuring mass

Qn. What is mass?

Mass is the amount of matter in an object.

Qn. State the basic unit for measuring mass.

Grammes

Qn. State the standard unit for measuring mass.

Kilogrammes

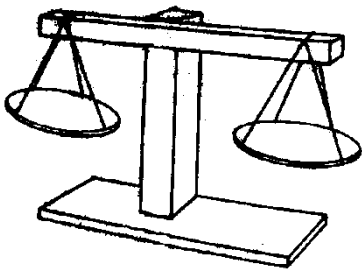
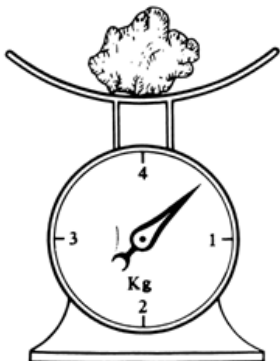
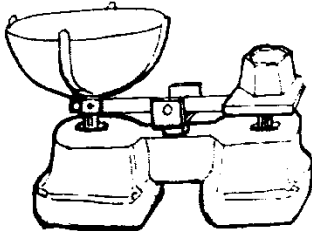
Qn. Identify any three instruments used to measure mass.

Beam balance

Clock balance

Lever balance

Illustration showing common weighing balance.

		
Beam balance	Clock balance	Lever balance

Measuring weight

Qn. What is weight?

Weight is a measure of how much force of gravity is acting on an object.

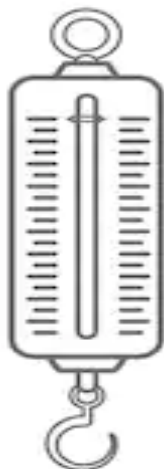
Qn. State the units in which weight is measured.

Newton

Qn. Name the instrument used to measure weight.

Spring balances

Illustration of spring balances



Qn. What is gravity?

Gravity is a force of attraction that pulls objects towards the centre of the earth.

Qn. Give a reason why objects thrown up come down on earth.

Due to gravitational force .

Qn. Name the force that objects thrown up come down on earth.

Gravity / gravitational force

Difference between weight and mass.

- (i) Mass is measured in grammes while weight is measured in Newtons.
- (ii) Mass is measured using a **beam balance** while weight is measured using a **spring balance**.
- (iii) Mass of an object does not change whenever it is measured while weight changes when measured in different places.
- (iv) Mass is the amount of matter in an object while weight is a measure of how much force of gravity is acting on an object.

Measuring capacity

Qn. What is capacity?

Capacity is the amount of liquid any container can hold.

Qn. State the basic unit for measuring capacity.

Litre

Qn. Mention any two examples of liquids people buy in litres.

Paraffin - cooking oil - water - milk

Qn. Name the instrument used to measure capacity of liquids.

Hydrometer

Measuring volume

Qn. What is volume?

Volume is the space occupied by an object.

Qn. State the units in which volume is measured.

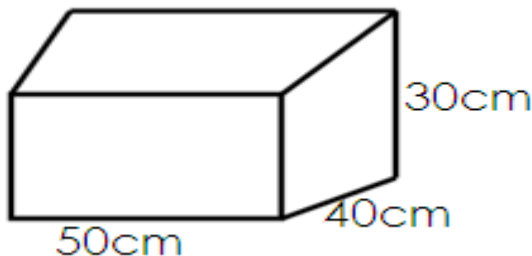
Cubic units

Finding capacity of a container.

To find the capacity of a container, first find its volume.

Example 1

Find the capacity of the container below.



To find volume we multiply length with the width and height.

$$\begin{aligned}\text{Volume} &= \text{Length} \times \text{Width} \times \text{Height} \\ &= 50\text{cm} \times 40\text{cm} \times 30\text{cm} \\ &= 6000\text{cm}^3\end{aligned}$$

To find the capacity of the container, we change the volume to litres.

$$1000\text{cm}^3 = 1 \text{ litre}$$

$$\begin{aligned}6000\text{cc} &= 6000 \div 1000 \\ &= 6 \text{ litres.}\end{aligned}$$

SHAPED OBJECTS

There are two groups of shaped objects namely;

- (a) Regular objects
- (b) Irregular objects

Regular objects

These are objects with well defined shapes.

Examples of regular objects.

A rectangular block

A cylindrical tank

A circular clock

Cuboids

Cube

Irregular objects

Irregular objects are objects that do not have a well defined shape.

Qn. Mention any two examples of irregular objects.

Stones

Broken glasses

Sweet potatoes

Irish potatoes

Mango

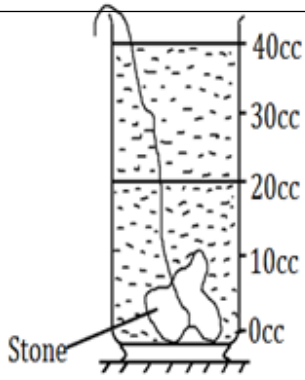
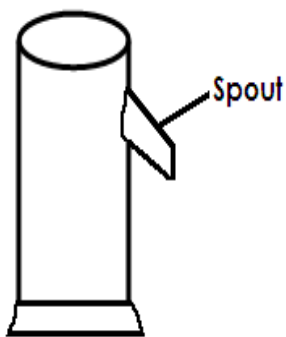
Measuring volume of irregular objects.

Qn. Name the method used to find the volume of irregular objects.

Displacement method.

Qn. Write down any two equipments use to find the volume of irregular objects.

- (i) Overflow can / Eureka can / Displacement can
- (ii) Measuring cylinder
- (iii) String

	
measuring cylinder	Eureka can

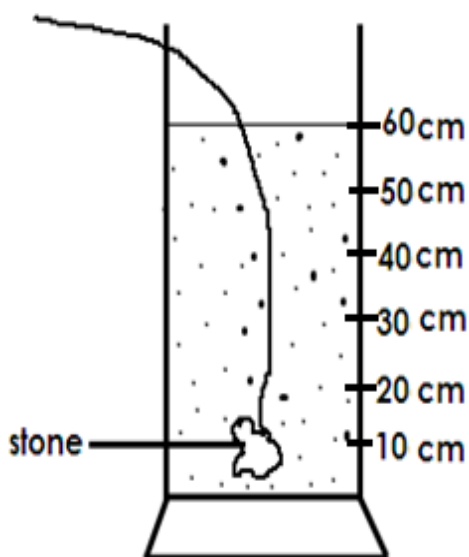
Note.

Measuring cylinder - Measuring the volume of an irregular object.

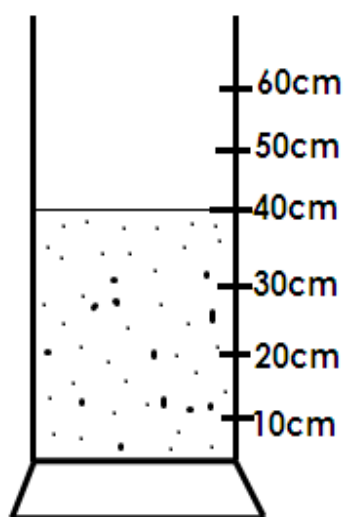
String - to lower the irregular object gently in water.

2. The experiment below was carried out by the P.5 class of ST Anthony primary school.

Study it carefully and use to answer the question that follows



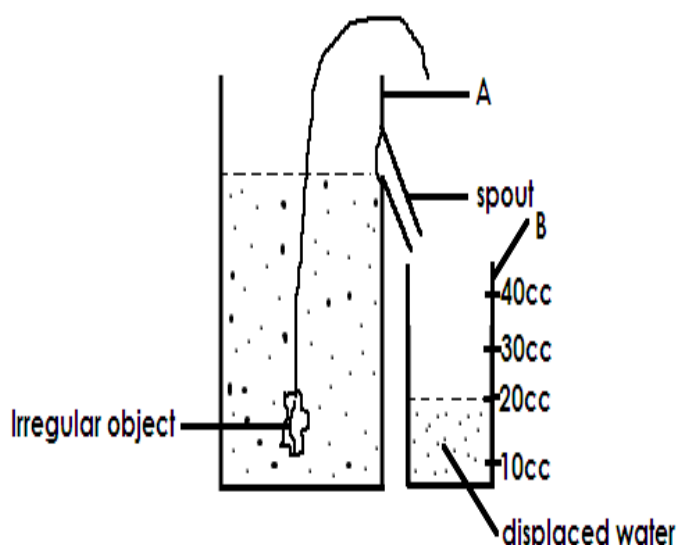
After lowering the stone



Before lowering the stone

$$\begin{aligned}
 \text{Volume} &= 2^{\text{nd}} \text{ level} - 1^{\text{st}} \text{ level} \\
 &= 60\text{cm} - 40\text{cm} \\
 &= \underline{20\text{cm}^3} \quad \text{or} \quad \underline{20\text{cc}}
 \end{aligned}$$

(b) Measuring the volume of an irregular object using a Eureka can and a measuring cylinder.



(i) Calculate the volume of the irregular object.

20cc

(ii) Give a reason for your answer above.

The volume of the water displaced is equal to the volume of the irregular object.

(iii) Name the container marked A and B.

A - Overflow can

B - Measuring cylinder

(iv) Why is the above method used when find the volume of an object like a stone?

A stone has no definite shape.

(v) When is the above method used?

When measuring the volume of an irregular object.

(vi) State the function of the measuring cylinder in the above experiment.

To measure the volume of displaced water.

DENSITY

Qn.What is density?

Density is mass per unit volume.

Qn. State the units in which density is measured.

Grammes per cubic centimeters or milliliters (gm / cc or gm /)

Finding density

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$D = \frac{M}{V}$$

Example I

An object has mass 24gm and volume 8cc. Find its density.

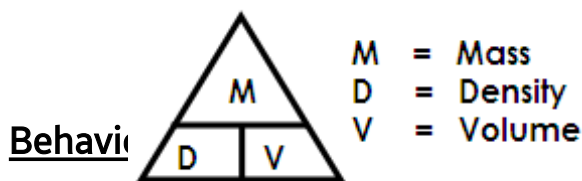
$$\begin{aligned} D &= \frac{M}{V} \\ &= \frac{24\text{gm}}{8\text{cc}} \\ &= 3\text{gm /cc} \end{aligned}$$

Activity

1. Find the mass of an irregular object with density 4gm / cc and volume 10cc.
2. Find the volume of the irregular object if its mass is 50gm and the density is 5gm / cc.

3. Calculate the density of the irregular object if its mass is 40gm and the volume is 20cc.
4. A brick has mass of 72gm and density of 6gm/cc. Calculate its volume.
5. Find the mass of an object whose density is 3gm / cc and volume 8 cc.

NOTE: You can use this triangle to remember the formulae



Qn. What happens to a leaf when put in water?

It floats on water

Qn. What happens to a coin of sh. 500 when put in water?

It sinks in water

Floating and sinking objects

Qn. What are floating objects?

These are objects which remain on top of water when put in it.

Qn. Why do some objects float on water?

Some objects have less density than water.

Examples of floating objects.

Leaf

A feather

A paper

Dry wood

Qn. Give a reason why a leaf floats on water.

The density of a leaf is less than that of water.

Qn. What are sinking objects?

These are objects which settle at the bottom of water when put in it

Qn. Why do some objects sink in water?

Some objects have more density than water.

Examples of sinking objects.

(i) A stone

(ii) A coin

(iii) Nail

(iv) Sand

Qn. Give a reason why a stone sinks in water.

The density of a stone is more than that of water.

Qn. Name the force that makes objects weigh less when put in water.

Up thrust force / buoyancy force

Qn. What is buoyancy force?

Is the force that makes objects weigh less when put in water.

Qn. Mention the instrument used to measure the density of liquids.

Hydrometer.

TOPIC 3

IMMUNITY AND IMMUNISATION

Qn. What is immunity?

Immunity is the ability of the body to fight against disease causing germs.

Importance of immunity.

Immunity enables the body to resist diseases.

Qn. What is immunization?

Immunisation is the introduction of vaccines into the body.

Qn. What are vaccines?

Vaccines are special drugs that are introduced into the body to make it produce antibodies against certain disease causing germs.

Types of vaccines

- (i) Toxoid vaccines e.g. tetanus toxoid.
- (ii) Attenuated living vaccines e.g. BCG vaccine, measles vaccine, Rubella vaccine, OPV vaccine
- (iii) Killed vaccines e.g. IPV vaccine, rabies vaccine, cholera vaccine, influenza vaccine, hepatitis A vaccine

Importance of vaccines

Vaccines enable the body to produce antibodies against certain disease causing germs.

Types of immunity

- (a) Natural immunity
- (b) Acquired artificial immunity

Qn. What is natural immunity?

This is the type of immunity where the body builds its own antibodies.

Qn. Give three ways how a baby can acquire natural immunity.

- (i) Through breast feeding.
- (ii) After recovering from an illness
- (iii) From a pregnant woman to her unborn child through the placenta.

Qn. Mention two diseases whose immunity can be built after recovering from it.

Measles
Mumps

Qn. What is artificial immunity?

Is the type of immunity got mainly through immunization.

Qn. State one way how the body can acquire artificial immunity.

Through immunization

Importance of immunization.

- (i) It boosts the immunity of the body.
- (ii) Immunisation enables that body to produce antibodies.
- (iii) Immunisation protects the body against childhood killer disease.
- (iv) Immunisation reduces infant mortality rate.

Qn. State any two methods how vaccines are introduced into the body.

Oral method

Injection method

CHILDHOOD OR INFANT KILLER DISEASES

Qn. Identify any four examples of immunisable childhood diseases.

Polio

Tuberculosis

Diphtheria

Hepatitis B

Tetanus

Whooping cough (pertussis)

Measles

Haemophilus influenzae type b

POLIOMYELITIS (POLIO)

Qn. What causes polio?

polio virus

Qn. How is polio spread?

Through drinking water contaminated with polio virus.

By house flies

Signs and symptoms of polio

Weakness of bones and muscles

Lameness

Paralysis of limbs

Fever

Prevention of polio

Drinking boiled water

Proper disposal of human faeces.

Immunizing all children at birth using polio vaccine.

Qn. How is the polio vaccine given to babies?

Orally or Through drops in the mouth.

TUBERCULOSIS

Qn. What causes tuberculosis?

Bacteria

Qn. How is tuberculosis spread?

Through contaminated air.

Signs and symptoms of tuberculosis

- (i) Chronic cough
- (ii) Loss of weight
- (iii) A lot of sweating at night
- (iv) Loss of appetite
- (v) Pain in the chest.

Prevention of tuberculosis

Immunizing children with the BCG vaccine

Always drink boiled milk

Isolation of an infected person.

DIPHTHERIA

It is caused by bacteria

It spreads through contaminated air.

Signs and symptoms of diphtheria

Swollen neck

Sore throat

Fever

Difficulty in breathing

Headache

Prevention of diphtheria

Immunizing children with the DPT vaccine.

(PERTUSSIS (WHOOPIING COUGH))

It is caused by bacteria

It spreads through air

Signs and symptom of pertussis

Runny nose

Severe coughing

Loss of weight

Wheezing sound while coughing

TETANUS

It is caused by bacteria.

Qn. How is tetanus spread?

Through dirty wounds and cuts.

At birth when cutting the umbilical cord with a contaminated object.

Signs and symptoms of tetanus

Stiff muscle all over the body.

Spasm when touched

The baby stops suckling

Fever and headache

Prevention of tetanus

Immunising children with DPT vaccine.

Pregnant women should be given (TT) Tetanus Toxoid vaccine.

Wounds and cuts should always be dressed and kept clean.

Use clean objects to cut the umbilical cord of a baby.

Qn. Why are pregnant women and girls of child bearing age given Tetanus toxoid vaccine?

To protect them and their unborn babies against tetanus.

Qn. Give a reason why DPT vaccine is called a triple vaccine?

It is given against three immunisable childhood diseases.

Qn. Why is polio vaccine given to babies at birth?

Babies are not born with immunity against polio.

Qn. Name the vaccine given to a baby once in life time.

BCG vaccine

HEPATITIS B

It is caused by virus.

Qn. Name the body organ which is affected by hepatitis B.

Liver

Qn. How is hepatitis B spread?

Through sharing sharp skin piercing objects.

Through blood transfusion with contaminated blood.

Through unprotected sexual intercourse with an infected person.

Signs and symptoms of hepatitis B

Dark urine

Greyish stool

Yellowish eyes

Vomiting

Tiredness

Fever

Prevention of hepatitis B

Immunise all children with Hepatitis B vaccine.

Abstain from sex.

Do not share sharp piercing objects.

Haemophilus influenza B.

It is caused by virus.

It is spread through air.

Signs and symptoms of haemophilus influenza b

Immunisation of children with Hib vaccine.

Treating the infected person in isolation.

MEASLES

It is caused by virus

It spreads through air.

Signs and symptoms of measles

Skin rash all over the body.

Sore mouth

Red eyes

Running nose.

Prevention of measles

Immunize children using measles vaccine.

Isolate infected children

Qn. Why is measles vaccine given to babies at nine months?

The babies are born with immunity against measles which weakens at 9 months.

Qn. What is a pentavalent vaccine?

Is a vaccine with five antigens.

Qn. Mention any two examples of a pentavalent vaccine.

DPT vaccine

Hep B vaccine

Hib vaccine

A table showing the child's immunization schedule.

DISEASE	VACCINE	AGE	METHOD OF ADMINISTRATION
Tuberculosis	BCG vaccine	At birth	by injection on the right upper arm.
Polio	Polio vaccine	At birth 6 weeks 10 weeks 14 weeks	Orally (drops in the mouth)
- Diphtheria	DPT – Hep B +	6 weeks	By injection on the left upper

- Pertussis - Tetanus - Hepatitis B - Haemophilus - influenza b	Hib	10 weeks 14 weeks	arm.
Measles	Measles vaccine	9 months	injection on the left upper arm.

Qn. Name the vaccines given to babies at birth.

BCG vaccine

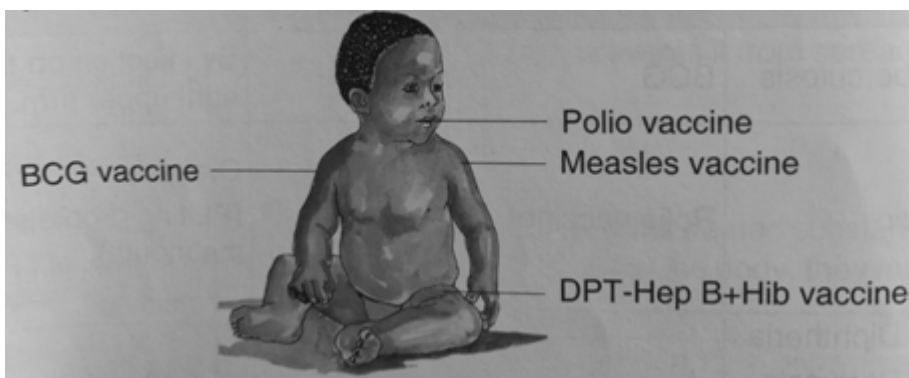
Polio vaccine

Qn. Mention two diseases whose vaccine are administered to babies at birth.

Polio

Tuberculosis

An illustration showing the immunization sites and vaccine.



Other immunisable diseases and their vaccines.

Cholera - oral cholera vaccine

Meningitis - Meningococcal vaccine

Yellow fever - Yellow fever vaccine

Rabies	-	Rabies vaccine
Rubella / German measles	-	Rubella vaccine
Pneumonia	-	Pneumococcal conjugate vaccine (PCV)
Diarrhoea	-	Rotavirus vaccine
Cancer of the cervix	-	HPV vaccine

CHILD HEALTH CARD

Is a card which contains the child's growth and health records.

Qn. Write down at least four important information found on a child health card.

Child's name

Child's sex

Child's date of birth

Child's birth order

Child's weight at birth

Mother's occupation

Father's name

Qn. State the importance of a child health card to a parent.

It helps the parent to monitor the growth of the child.

It helps the parent to know the date of the next immunization.

Qn. How is a child health card important to a health worker?

It helps the health worker to know which dose of immunization was given and what is remaining.

Qn. How can a P.5 teacher tell that a child was immunized against tuberculosis without using a child health card?

Checking for the immunization scar on the right upper arm of the child.

Qn. How is a child health card useful to a school?

It helps the school management to know if the child was immunized or not.

Qn. How can a P.5 pupil participate in the immunization programme?

Reminding the parent of the dates of immunization.

Taking the siblings for immunization.

Singing songs that interest people about immunization.

Acting plays about immunization.

Organizing the immunization centres.

Qn. State one way a family can participate in the immunization programmes.

Taking all children in a family for immunization.

Qn. State one way a community can participate in the immunization programme.

Setting up immunization centres.

Qn. How can the government of Uganda promote immunization programme?

Providing vaccines

Building health centres

Training health workers

Qn. Why is immunization free in Uganda?

To enable all parents take their children for immunization.

Qn. Write the following in full.

UNEPI - Uganda National Expanded Programme on Immunisation.

NIDs - National Immunisation Days

Qn. Suggest the roles played by UNEPI

To supply vaccines to different health centres.

To teach people the importance of immunization.

Training health workers on how to carry out immunization.

Qn. Name the special refrigerator in which vaccines are stored.

Vaccine carrier

Qn. Why are vaccines stored cool conditions.

To prevent them from going bad easily

A child health card

REPUBLIC OF UGANDA



MINISTRY OF HEALTH

CHILD HEALTH CARD

District:		Child's Registration No.:
Health Unit:		
Child's Name:		Birth Weight(Kg):
Sex:	Date of Birth: / /	Birth Order:
Mother's Name:		Mother's Occupation:
Father's Name:		Father's Occupation:
Where the Child lives:		
Sub County/Division:		
Parish:		
L.C.I:		

TICK REASONS FOR SPECIAL CARE:			
Birth weight less than 2.5 kg		Brothers or Sisters undernourished	
Birth defect		Mother dead	
Other handicaps or illness		Father dead	
Fifth child or more		3 or more children in family dead	
Birth less than 2yrs after last birth		Twin child	

ANY OTHER REASON FOR SPECIAL ATTENTION:

THE DIGESTIVE SYSTEM

Qn. What is the digestive system?

Please carry this card every time you bring your child for care or attention.

Is a group of organs which help to break down food making it usable by the body.

Qn. What is digestion?

Is the process by which food is broken down into smaller soluble particles that can be absorbed into the blood stream.

Qn. What is an alimentary canal?

Is a muscular tube running from the mouth to the anus.

Qn. Mention the parts that make up the alimentary canal.

Mouth

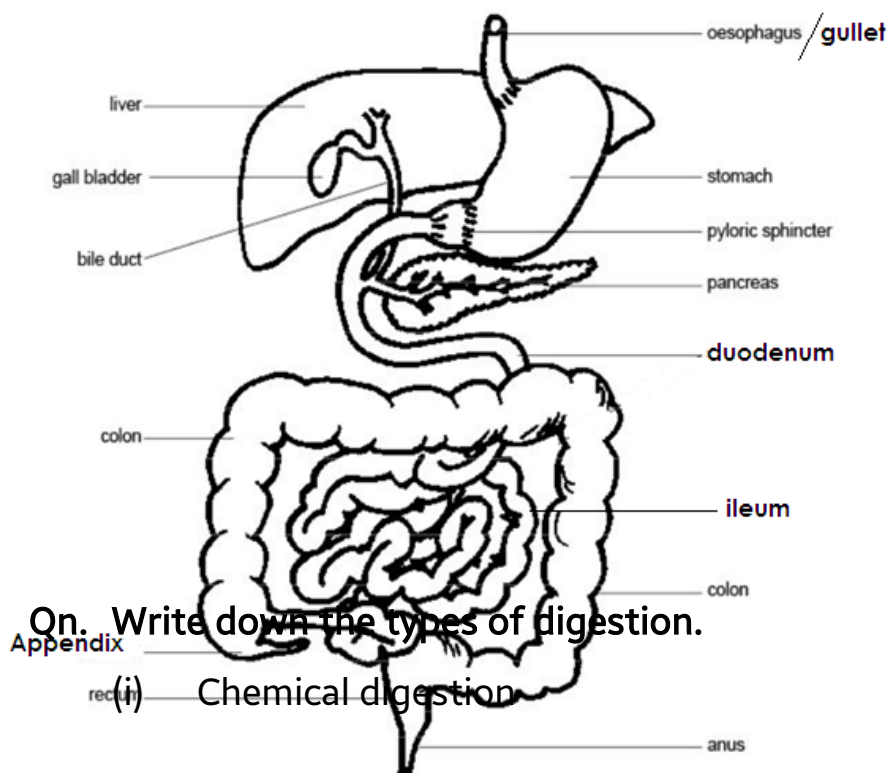
Gullet

Stomach

Small intestine

Large intestine

Diagram showing the digestive system.



Qn. Write down the types of digestion.

(i) Chemical digestion

(ii) Mechanical digestion/ physical digestion

Qn. What is mechanical / physical digestion?

Is the type digestion carried out by the action of the teeth in the mouth.

Qn. What chemical digestion?

Is the type of digestion carried out by enzymes.

ENZYMES

Qn. What are enzymes?

Enzymes are chemical substances that speed up the digestion of food.

Qn. State any four characteristics of enzymes.

They are destroyed by heat/ high temperature.

They are proteins in nature.

They always form the same end product.

They act on one kind of food.

Digestion in the mouth

Qn. Where does digestion of food in man begin?

In the mouth

Qn. Where in man does digestion of food end?

In the ileum

Qn. State the role of the following in the mouth during digestion.

Teeth - To break food into small particles.

Tongue - To roll food into a bolus.

Salivary glands -To produce saliva

Saliva -Softens down food

Qn. Name one enzyme found in saliva.

Salivary amylase (ptyalin)

Qn. Under what condition does salivary amylase work?

Alkaline condition

Qn. State the use of salivary amylase in the mouth.

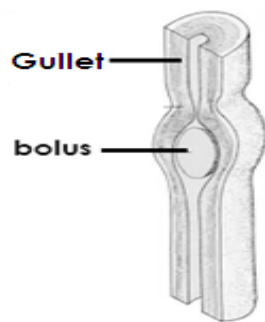
Digests cooked starch in the mouth

Qn. What is peristalsis?

Peristalsis is the wave-like movement of food through the alimentary canal.

Movement of substances through tubular organ.

Diagram showing peristalsis



Qn. Give the func

Passes food from the mouth to the stomach.

DIGESTION IN THE STOMACH

The stomach stores food for a short period of time.

Qn. Name the gland found in the stomach.

Gastric gland

Qn. State the function of the gastric gland.

To produce gastric juice.

Qn. Mention the two enzymes contained in gastric juice.

Pepsin

Rennin

Qn. State the function of each of the following enzymes in the stomach.

Pepsin - Digests proteins

Rennin - Clots milk protein in babies.

Qn. Name the enzymes found in the stomach of babies.

Rennin

Qn. Name the acid produced by the stomach walls.

Hydrochloric acid

Qn. State the function of hydrochloric acid found in the stomach.

To kill germs that come along with food.

To provide favourable conditions for enzymes in the stomach to act upon food.

Qn. Under what condition do pepsin and Rennin work?

Acidic condition

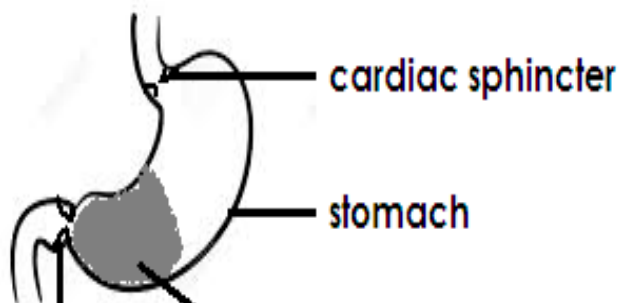
Qn. What is churning?

Is the process by which food is mixed with digestive juices in the stomach.

Qn. What name is given to the food which has been churned in the stomach.

Chyme

The stomach



Qn. State the function of each of the following sphincter muscle.

(a) Cardiac sphincter muscles

pyloric sphincter

To allow little food from the gullet to the stomach at a time.

(b) Pyloric sphincter muscle

To allow a little chyme to the duodenum at a time.

Qn. Mention any two substances that are absorbed in the stomach.

Alcohol

Common salt

Simple sugars

Medicine

Digestion in the duodenum

Qn. Write down three enzymes found in the duodenum.

Lipase

Amylase

Trypsin

Qn. State the function of the pancreas.

To produce pancreatic juice.

To produce insulin hormone

Qn. State the function of the following pancreatic enzymes.

(i) **Lipase** - It changes fats into fatty acids and glycerol.

(ii) **Amylase** - It changes uncooked starch into maltose.

(iii) **Trypsin** - Changes peptides into amino acids.

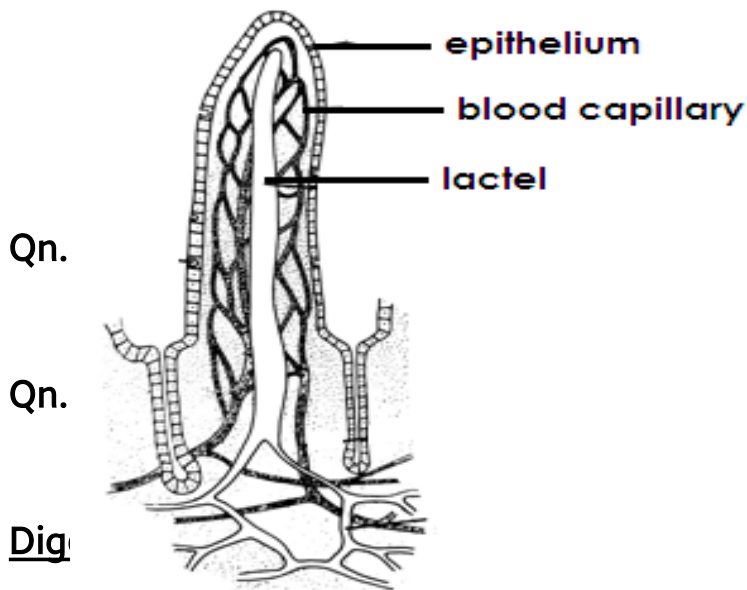
Qn. State the functions of the liver.

It produces bile juice

Stores important vitamins

Stores important mineral salts.

To control blood sugars



Digestion of food ends in the ileum.

Qn. Name the digestive juice produced in the ileum

Succus entericus

Qn. Write down three enzymes found in the ileum.

Lactose - Peptidase - Galactase - Maltase

Sucrase

Qn. Identify the importance process that takes place in the ileum.

Absorption of digested food.

Qn. State two ways how the ileum is adapted to its function.

It has the villi which help in food absorption.

It is fairly long and coiled which increases the surface area for food absorption.

Qn. Name the finger like structure that helps in the absorption of food in the ileum.

Villi

Structure of the villus

Qn. How are the villi adapted to their function?

They are thick walled for easy diffusion of food.

The villi are surrounded by a great network of blood capillaries.

Qn. Mention two parts that make up the small intestines.

Duodenum

Ileum

Large intestines

Qn. Outline any three parts that make up the large intestines.

Colon

Rectum

Anus

Qn. State one important process that takes place in the large intestines.

Absorption of water.

Qn. Give a reason why digestion of food cannot take place in the colon.

There is no enzyme in the colon to digest food

Qn. Why can't absorption of digested food take place in the colon?

There is no villi in the colon

Qn. State the function of the appendix.

Stores undigested materials like stones

Qn. What is absorption?

Is the process by which digested food is taken into the blood stream.

Qn. By what process is digested food absorbed into the blood stream in the ileum?

Diffusion

Qn. Give the function of the rectum.

Stores faeces before they are passed out.

Qn. Mention the end products of digestion of each of the following food values.

(a) **Proteins** - Amino acids

(b) **Carbohydrates** - Glucose

(c) **Fats** - fatty acids and glycerol

Qn. Give a reason why vitamins and mineral salts are not digested.

Vitamins and mineral salts are soluble.

Qn. Name one blood vessel that transports blood rich in digested food from the ileum to the liver.

Hepatic portal vein

Diseases of the digestive system

Cholera - Diarrhoea - Peptic ulcers - Typhoid

Hepatitis B - Dysentery - Appendicitis

Disorder of the digestive system

Vomiting - Constipation

Intestinal obstruction - Indigestion

Qn. State any two cause of constipation

Shortage of water.

Shortage roughages.

Qn. State two ways of preventing constipation.

Eat fruits and vegetables regularly.

Drink water or juice after eating food.

Eat food rich in roughage.

Qn. Which disorder of the digestive system is Okello who eats hurriedly without chewing food properly likely to face?

Indigestion

Qn. State any two ways a P.5 pupil can avoid indigestion.

Do not eat too much.

Chew food properly before swallowing.

Drink water after eating food.

Ways of preventing diseases of the digestive system.

Drinking boiled water

Washing hands before handling food.

Washing hands after visiting a latrine or toilet.

Ways of keeping the digestive system in good health working conditions.

Maintaining good eating habits.

Eat a balanced diet.

Doing physical exercises daily.

Drinking clean boiled water.

I WISH YOU GOOD LUCK